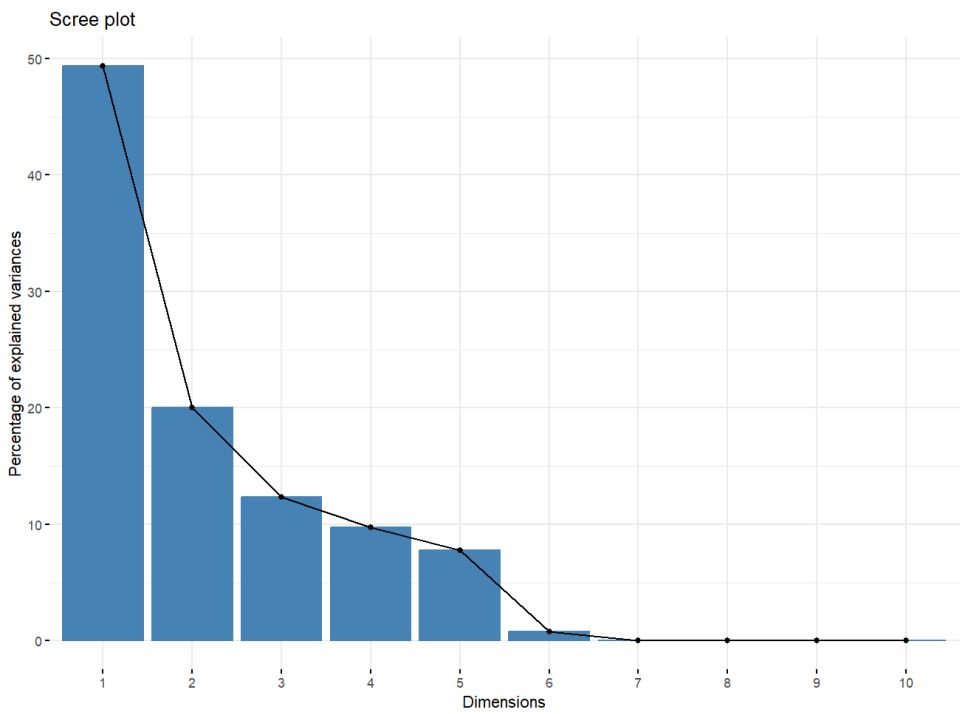
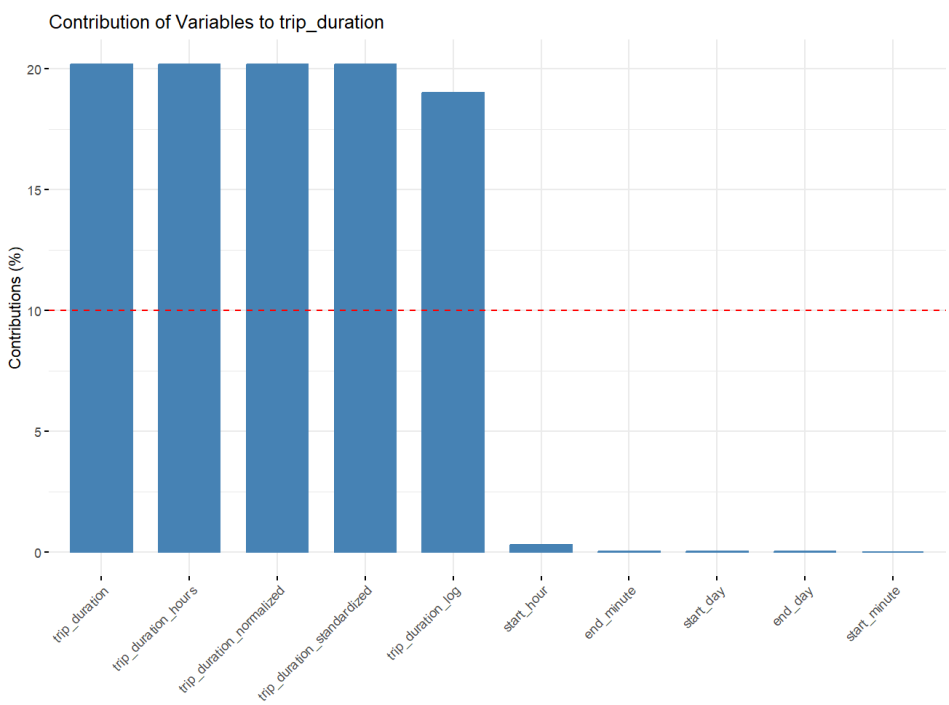
Scree Plot:

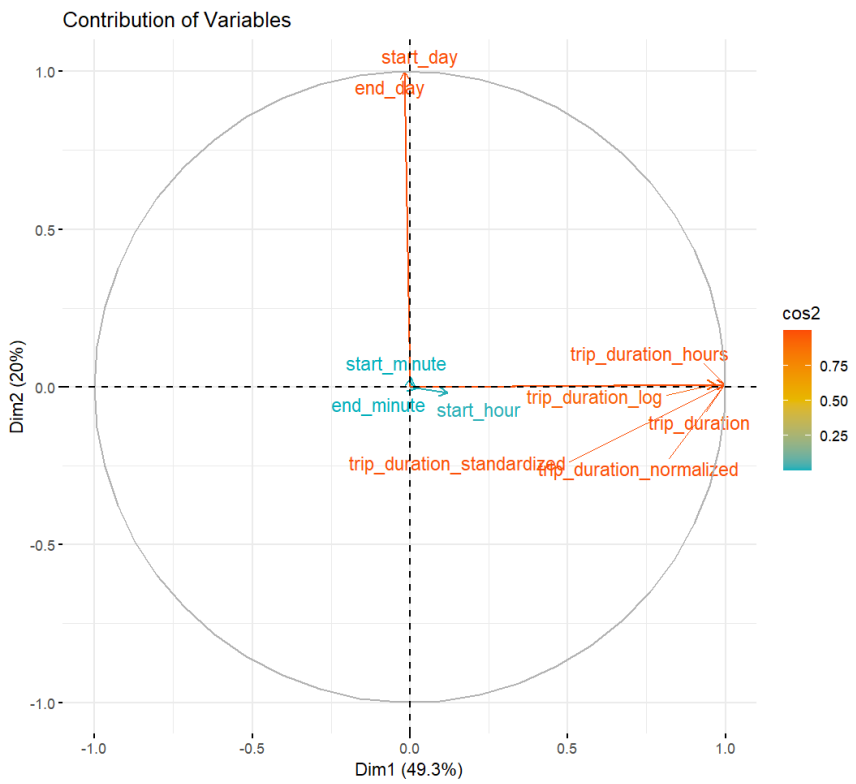


- The scree plot is used to determine the number of principal components to retain where the x-axis shows dimensions representing the principal components. In contrast, the y-axis represents the percentage of total variance in the data for each component. Assume that the first component is trip_duration then it has the highest variance in the data of nearly 50%.

Contribution of variables towards trip_duration:

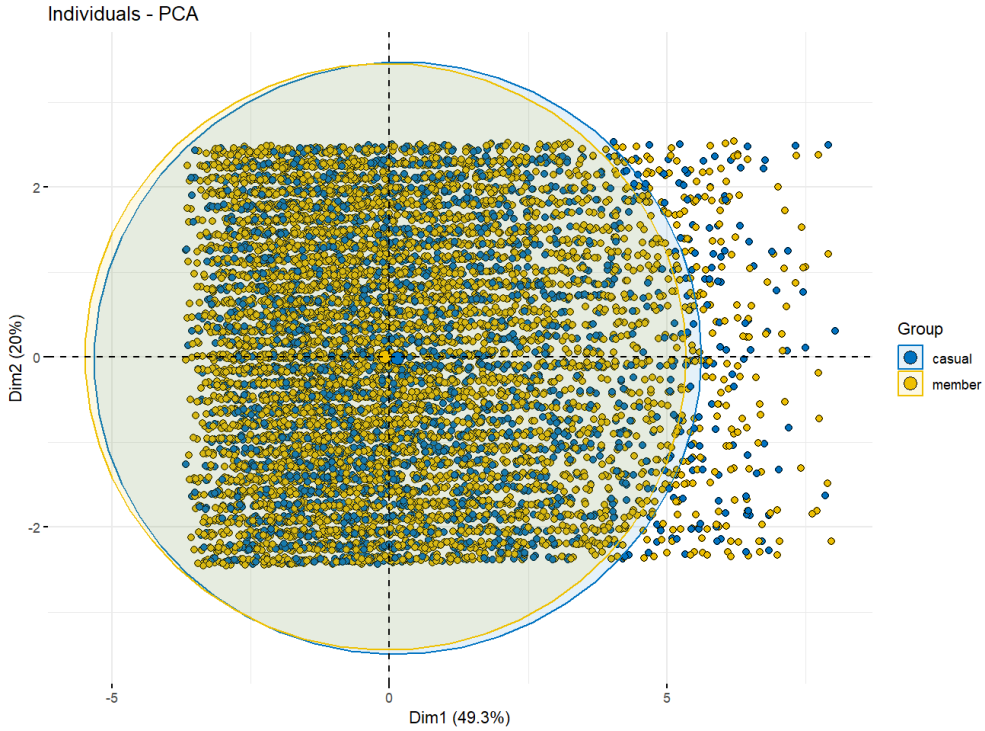


- This bar chart shows how different variables contribute to the duration of a trip where the x-axis represents the variables that affect trip duration whereas the y-axis represents the percentage contribution of each of the variables. The variables trip_duration, trip_duration_hours, trip_duration_ normalized, trip_duration_standardized, trip_duration_log has the highest percentage of contributing towards trip duration.



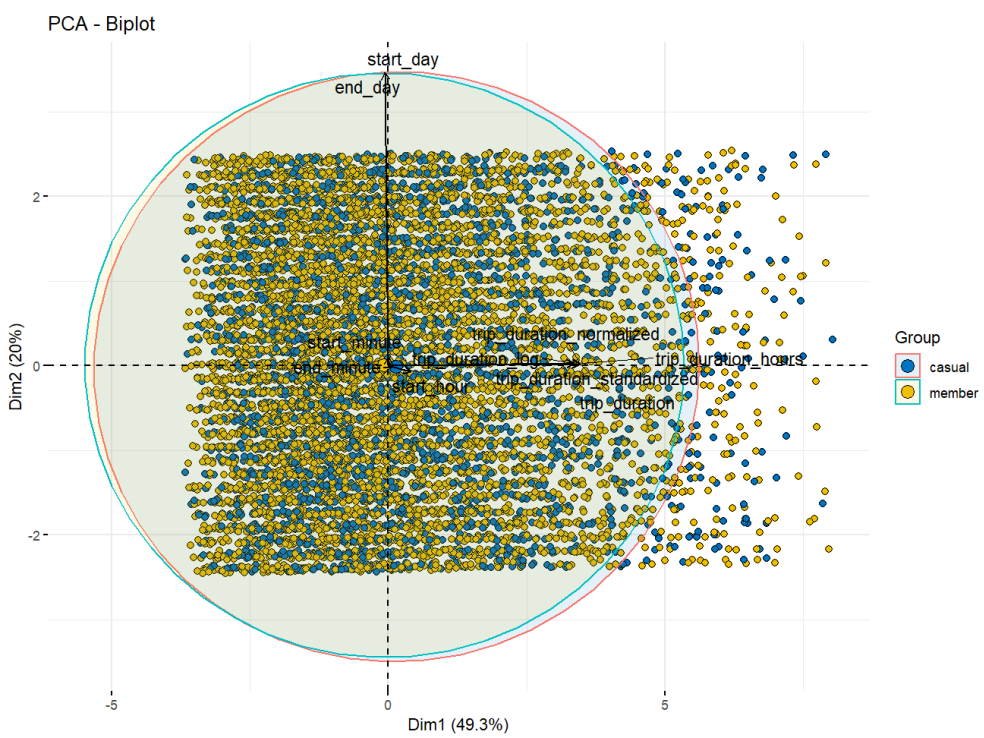
- This biplot visualizes the contributions of various variables to two principal components. The axis represents the first two principal components, which together explain a significant portion of the variance in the data (49.3% for Dim1 and 20% for Dim2). Variables like trip_duration_hours and trip_duration_log have strong contributions with Dim1, indicating they explain a lot of the variance captured by this component whereas variables like start_day and end_day contribute more to Dim2, suggesting they explain variance captured by this component.

Individual PCA Biplot:



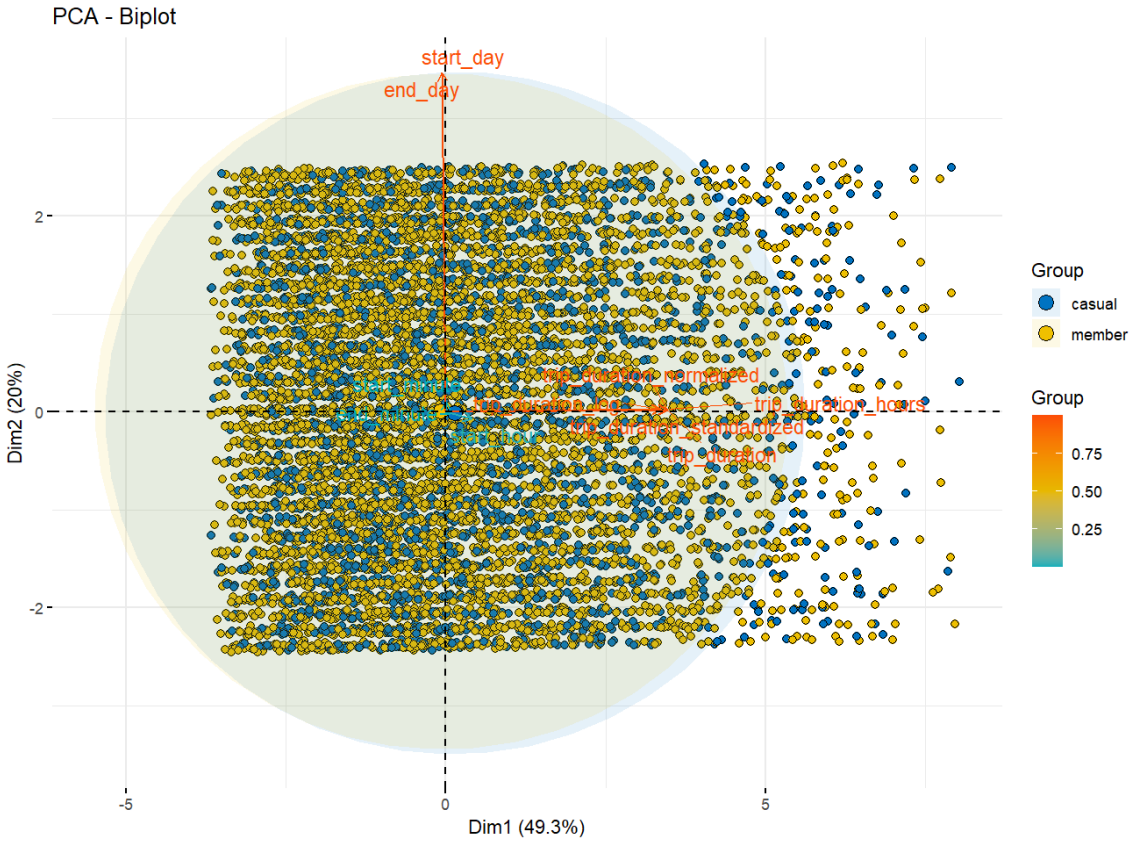
- This scatter plot shows two components on each axis with 49.3% of the variance on Dim1 compared to 26.9% on Dim2 of the variance in the data that captures a significant portion of the variability in the dataset. Each point represents an individual observation, and they are based on two groups: casual and member. Points that are close to each other on the plot are similar in terms of the principal components whereas the points that are far apart are dissimilar and the separation between the yellow and blue points indicates differences between the casual and member groups.

Combined PCA Biplot:



- This scatter plot shows two components on each axis with 49.3% of the variance on Dim1 compared to 26.9% on Dim2 of the variance in the data that captures a significant portion of the variability in the dataset. Each point represents an individual observation, and they are based on two groups: casual and member. The labeled vectors indicate the direction and magnitude of each variable's contribution to the principal components such as variables like start_day and end_day have arrows pointing in specific directions, showing how they contribute to Dim1 and Dim2. The green ellipse encompasses most of the data points and shows a cluster for both groups which indicates that the majority of the data points share similar characteristics. Points that are close to each other on the plot are similar in terms of the principal components whereas the points that are far apart are dissimilar and the separation between the yellow and blue points indicates differences between the casual and member groups.

Combined PCA Biplot with cos2:



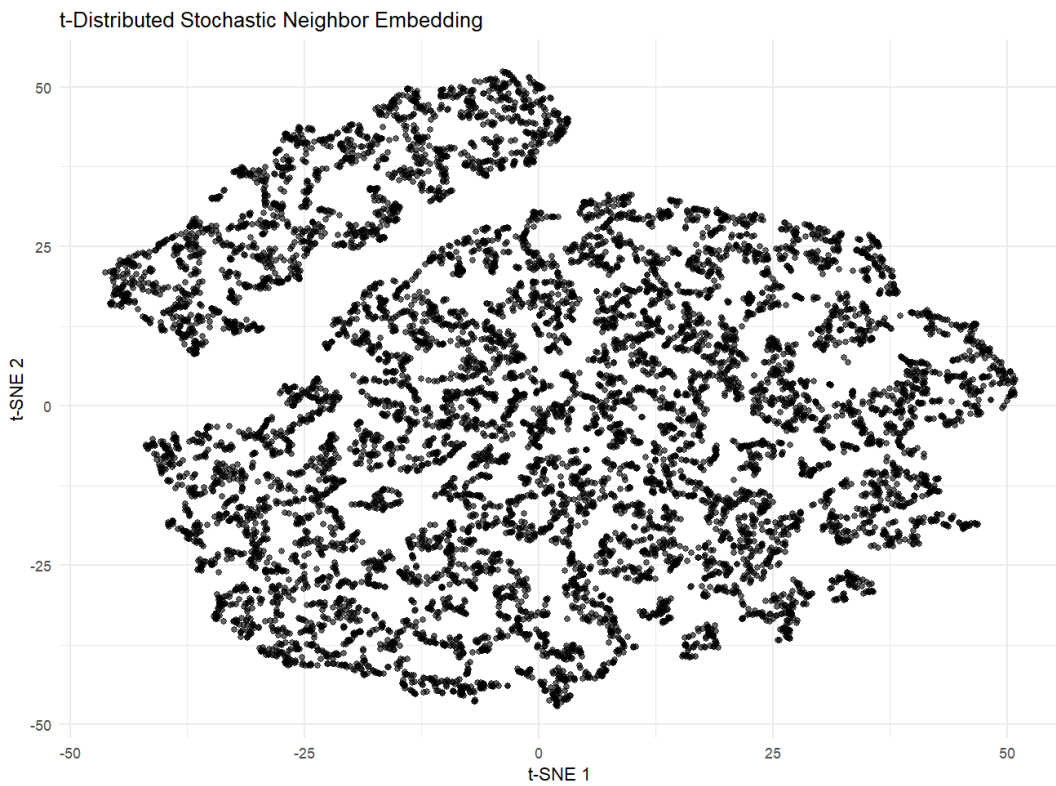
This scatter plot shows two components on each axis with 49.3% of the variance on Dim1 compared to 26.9% on Dim2 of the variance in the data that captures a significant portion of the variability in the dataset. Each point represents an individual observation, they are based on two groups: casual and member where the shading of the points indicates the value of another variable ranging from 0.25 to 0.75. The labeled vectors indicate the direction and magnitude of each variable's contribution to the principal components such as variables like start_day and end_day have arrows pointing in specific directions, showing how they contribute to Dim1 and Dim2. The green ellipse encompasses most of the data points and shows a cluster for both groups which indicates that the majority of the data points share similar characteristics. Points that are close to each other on the plot are similar in terms of the principal components whereas the points that are far apart are dissimilar and the separation between the yellow and blue points indicates differences between the casual and member groups.

Additional PCA Plots:

- This scatter plot visualizes the relationship between trip duration (in hours) and the hour of the day when trips start where The x-axis represents trip_duration_hours which is the duration of trips in hours and the y-axis represents start_hour which is the hour of the day when trips start. Each point represents an individual trip and the points are color-coded into two groups: casual users and members. The points are densely plotted, creating a cloud-like distribution across the graph but there are no immediately discernible patterns or clusters, indicating that trip duration and start hour are spread out across the dataset.

- This scatter plot visualizes the relationship between the start day and end day of trips where the x-axis represents start_day which is the day when trips start whereas the y-axis represents end_day which is the day when trips end. Each point represents an individual trip and the points are color-coded into two groups: casual users and members. Points that are close to each other on the plot are similar in terms of the principal components whereas points that are far apart are dissimilar and the separation between the red and blue points indicates differences between the two groups.

t-sne plot:



- This scatter plot visualizes high-dimensional data in a two-dimensional space where each point represents an individual observation and the points are distributed across the plot forming two distinct clusters. The presence of two distinct clusters suggests that the t-SNE algorithm has identified two groups within the data based on their similarity where one cluster is larger than the other which indicates a difference in the number of observations within each group. Points that are close to each other on the plot are similar in terms of the high-dimensional data whereas points that are far apart are dissimilar.